

PrimeIMU: High-Frequency Video-to-IMU Synthesis via Physics-Guided Simulation and Hybrid U-Net Refinement

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Inertial Measurement Units (IMUs) are fundamental sensing components across robotics, drones, and other embodied systems, providing high-frequency inertial signals essential for control, stabilization, state estimation, and dynamics modeling. A central challenge, however, lies in collecting large-scale, high-frequency IMU datasets with reliable ground-truth labels, which are crucial for initial model training and systematic experimentation. In practice, deploying, calibrating, and synchronizing physical IMU sensors is costly, time-consuming, and often constrained by hardware limitations and operational requirements. To alleviate these challenges, recent work has explored synthesizing IMU signals from either model-synthesized motions or vision-derived kinematic estimates. While text-to-motion-based IMU synthesis approaches scale well, they lack real-world visual grounding. Vision-driven video-to-IMU approaches offer such grounding but are fundamentally constrained by the limited temporal resolution of visual inputs and the inherent mismatch between kinematic estimates and inertial measurements.

Building on these observations, we propose PrimeIMU, a framework that synthesizes IMU signals from vision-derived kinematic trajectories and physics-simulated IMU signals, using a hybrid U-Net that learns to map simulated signals to real sensor measurements. Extensive experiments demonstrate that PrimeIMU (1) improves the quality of inertial signal synthesis, (2) generalizes to unseen motion patterns while preserving realistic signal characteristics, (3) enables synthetic-only training of downstream models with performance close to real-sensor training and (4) yields further gains when used to augment real data, and (5) adapts across different datasets and sensor configurations with minimal fine-tuning, supporting downstream tasks under domain shift.

CCS Concepts: • **Computing methodologies** → **Machine learning**.

Additional Key Words and Phrases: IMU Signal Synthesis

ACM Reference Format:

Xingjian Diao, Heba Aly, Cici Chen, and Hongda Mao. 2018. PrimeIMU: High-Frequency Video-to-IMU Synthesis via Physics-Guided Simulation and Hybrid U-Net Refinement. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 24 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Inertial Measurement Unit (IMU) data are widely used across articulated mechanical and robotic systems [18, 19, 25, 34], including aerial drones [7, 26] and other embodied platforms [1, 8]. High-frequency inertial measurements capture

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Manuscript submitted to ACM

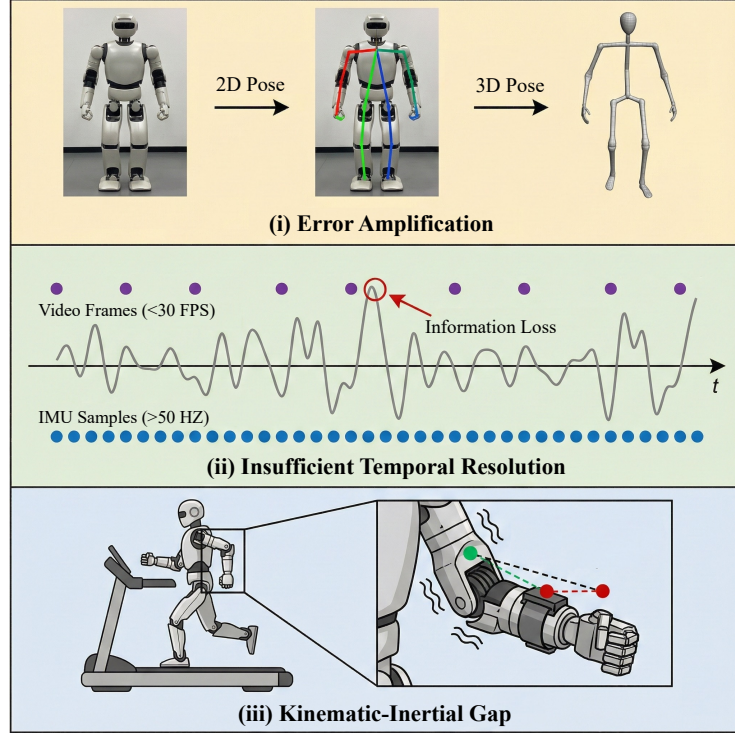


Fig. 1. Illustration of three fundamental limitations in current vision-driven IMU synthesis pipelines: (i) error amplification introduced by kinematic estimation, (ii) insufficient temporal resolution when projecting low-frame-rate video into high-frequency IMU signals, and (iii) discrepancies between idealized kinematic models and the motion sensed by an IMU mounted on a moving structure.

fine-grained motion dynamics [19, 34], supporting tasks such as motion analysis, dynamic behavior characterization [25], operational assessment [38], and model-based simulation of articulated structures [24]. Achieving these applications at scale requires large, high-frequency IMU datasets with reliable annotations. However, collecting such data remains highly challenging, as the process is time-consuming, costly, and requires substantial domain expertise [5, 11, 15, 23, 27, 36, 39].

To address the scarcity of labeled IMU data, prior work has explored synthetic IMU generation for data augmentation through two primary paradigms. The first paradigm leverages *large language models* (LLMs) to generate diverse motion descriptions, which are converted into 3D articulated trajectories via motion synthesis and subsequently transformed into virtual IMU signals through motion-to-IMU modeling [15, 16, 41]. The second paradigm relies on *vision-driven models* to extract 2D kinematic cues from visual inputs, lift them to 3D articulated representations, and synthesize virtual IMU signals via kinematic modeling [13, 14, 29]. While LLM-based approaches offer strong scalability, they lack visual grounding and often produce IMU signals with unrealistic dynamics. Vision-driven methods provide visual grounding, but still fall short of real IMU quality for three fundamental reasons (Figure 1). *First*, errors in visual kinematic estimation propagate and amplify through temporal differentiation, resulting in noisy or spiky inertial signals. *Second*, visual frame rates (typically 15–30 FPS) are substantially lower than IMU sampling rates ($> 50\text{Hz}$), leading to irreversible temporal under-resolution that cannot be recovered through interpolation. *Third*, a persistent kinematic–inertial gap exists between idealized articulated motion and the inertial response of physical IMU hardware, whose measurements

IMU Generation Method	Realism-Oriented Designs				
	real.	vis.+phys.	temp.res.	dev.adpt.	kin.t.freq.
IMUGPT [16]	✗	✗	✗	✗	✗
IMUGPT 2.0 [15]	✗	✗	✗	✗	✗
UniMTS [41]	✗	✗	✗	✗	✗
Video2IMU [14]	✗	✗	✗	✗	✗
Vi2IMU [29]	✗	✓	✗	✗	✗
IMUTube [13]	✗	✓	✗	✗	✗
PrimeIMU	✓	✓	✓	✓	✓

Table 1. Comparison of PrimeIMU with State-of-the-Art IMU generation methods on realism-oriented designs, where *real.*: explicitly aims for realistic IMU generation (beyond pure augmentation); *vis.+phys.*: combines visual grounding with physical modeling; *temp.res.*: addresses limited temporal resolution from video frame rates; *dev.adpt.*: adapts synthesis to device-specific characteristics (e.g., FPS, IMU sampling rates); *kin.t.freq.*: incorporates kinematic, temporal, and frequency-domain information during synthesis.

reflect mounting dynamics and other sensor-specific non-idealities. Together, these limitations substantially reduce the realism of synthesized IMU signals and limit the effectiveness of synthetic data for downstream learning tasks.

Addressing these challenges would provide significant value for both research and real-world deployment, enabling the development of IMU-driven models with minimal reliance on large-scale real sensor data. Consequently, one crucial question has arisen: *Is it feasible to synthesize IMU signals that faithfully capture the physical behavior of real sensors while remaining effective for IMU-based downstream applications?*

To address this question, we propose PrimeIMU, a framework for synthesizing realistic IMU signals from video-derived kinematics that explicitly targets key realism-oriented design challenges, including physical consistency, temporal resolution mismatch, and device-specific sensor behavior (Table 1). PrimeIMU first performs physics-based IMU simulation on video-derived kinematic trajectories and then refines the simulated signals using a hybrid U-Net to bridge the gap between idealized kinematics and real sensor measurements. Our contributions are as follows:

- **Realistic, Sensor-Faithful Video-to-IMU Synthesis.** We propose PrimeIMU, a video-driven IMU synthesis pipeline for generating realistic and sensor-faithful inertial signals. PrimeIMU integrates physics-based IMU simulation with a hybrid U-Net refinement module, jointly addressing three fundamental limitations of prior approaches: noise amplification from pose-derived kinematics, temporal under-resolution caused by low video frame rates, and the mismatch between idealized kinematic modeling and the behavior of physical IMU sensors.
- **Generalization to Unseen Motion Patterns.** We demonstrate that PrimeIMU generalizes to motion patterns not observed during training, suggesting that it captures underlying inertial structure beyond the training distribution.
- **Synthetic-Only Training of IMU-Based Models.** We show that downstream models trained solely on PrimeIMU-generated IMU signals achieve performance comparable to models trained on real sensor data, indicating that high-fidelity synthetic IMUs can substitute for real measurements when sensor data are limited.
- **Augmentation Benefits with Real IMU Data.** PrimeIMU-generated signals introduce sensor-faithful inertial patterns that complement real sensor measurements by expanding physically plausible inertial variations, leading to consistent performance gains in downstream tasks when used for data augmentation.
- **Cross-Dataset and Cross-Device Adaptation.** PrimeIMU adapts effectively to domain shifts across datasets and IMU hardware, requiring minimal fine-tuning to adapt to new sensor characteristics and sampling configurations.

2 Related Work

2.1 Data Scarcity and Learning Paradigms

Despite the widespread deployment of IMUs in articulated mechanical and robotic systems, the availability of large-scale, diverse, and well-calibrated inertial datasets remains limited. Constructing datasets that span a broad range of motion patterns, operating conditions, and device configurations requires controlled hardware deployment, precise calibration, and repeated data collection. These requirements impose substantial operational overhead and often lead to datasets that are narrow in scope and insufficiently diverse for robust inertial modeling. Moreover, variations in device characteristics and sensor mounting conditions introduce nontrivial device-dependent distribution shifts, further hindering cross-system generalization [10, 17]. As a result, existing IMU datasets remain constrained in operational coverage despite the ubiquity of micro-electro-mechanical systems (MEMS)-based sensors.

Beyond data quantity, annotation quality presents an additional bottleneck. Establishing reliable ground truth for IMU data typically relies on structured collection protocols or manual post-processing of continuous recordings, which can introduce inconsistencies in activity boundaries, temporal alignment, and labeling conventions [5, 11, 15, 23, 27, 36, 39]. Such variability further limits the scalability and uniformity of supervised IMU training corpora.

To mitigate these challenges, prior work has explored alternative learning paradigms, including self-supervised pretraining [12, 28], few-shot learning and domain adaptation [3, 9], metric-based representations [6], adversarial robustness techniques [2], and cross-dataset transfer [30]. While these approaches improve generalization under data scarcity, they remain fundamentally constrained by the diversity and physical coverage of real IMU measurements.

This limitation motivates the development of realistic synthetic IMU generation, where synthesized signals explicitly aim to reproduce device-level inertial behavior while scaling across motion types and hardware configurations. Such synthetic data can complement limited real measurements and enable effective downstream inertial modeling when large-scale physical data collection is impractical.

2.2 Cross-Modality IMU Synthesis

Owing to the difficulty of collecting large-scale annotated IMU datasets, recent work has increasingly explored the synthesis of inertial signals from alternative sensing modalities. Existing approaches broadly fall into two categories: text-driven and video-driven synthesis.

Text-driven pipelines, such as IMUGPT [16] and IMUGPT 2.0 [15], leverage large language models to generate diverse motion descriptions, which are then mapped to parametric trajectories and converted into synthetic IMU sequences. UniMTS [41] further incorporates linguistic priors before translating text descriptions into skeletal motions. While these approaches offer strong scalability and diversity, they lack visual grounding and often struggle to preserve the statistical and physical characteristics of real IMU measurements.

Video-driven pipelines introduce visual grounding by extracting motion cues directly from videos. IMUTube [13] estimates 2D poses, lifts them to 3D, and differentiates the resulting trajectories to approximate inertial signals. Video2IMU [14] directly regresses IMU sequences from 2D pose representations, while Vi2IMU [29] focuses on wrist-mounted sensors in constrained communication scenarios. Compared to text-driven methods, video-driven approaches generally yield more realistic inertial signatures due to their explicit visual grounding.

Despite these advantages, video-driven IMU synthesis remains fundamentally limited. Errors in 2D and 3D pose estimation are amplified through temporal differentiation, resulting in unstable or noisy inertial predictions. In addition,

the inherent mismatch between video frame rates and IMU sampling frequencies constrains the synthesis of high-frequency motion components. More importantly, visually derived kinematic trajectories alone cannot fully capture the inertial dynamics measured by physical IMU sensors, whose outputs depend on forces, vibrations, and mounting-dependent effects that are not represented in purely kinematic models. Together, these limitations indicate that effective IMU synthesis requires jointly leveraging visual kinematics and physics-based inertial modeling to more faithfully approximate the behavior of real IMU sensors.

3 Method

Our objective is to synthesize high-frequency and physically plausible inertial measurement unit (IMU) signals from standard RGB video inputs. This problem is challenging due to the substantial domain gap between low-frame-rate visual observations and high-frequency inertial measurements, as well as the noise and estimation errors inherent in vision-based motion capture.

To address these challenges, we propose PrimeIMU, a two-stage synthesis framework illustrated in Figure 2. In the first stage, visual motion is estimated from RGB videos and translated into an initial IMU signal using physics-based kinematic modeling. This stage captures coarse motion trends but remains limited by low temporal resolution and accumulated estimation noise. In the second stage, a hybrid U-Net-based generative refinement module reconstructs high-frequency inertial dynamics, suppresses noise, and enforces kinematic consistency at the target IMU sampling rate, producing realistic and sensor-faithful IMU signals.

3.1 Stage 1: Physics-Guided IMU Simulation

The first stage of PrimeIMU converts the input video into a coarse IMU estimate by extracting 3D kinematic trajectories and applying a physics-guided inertial simulation model. This stage provides a physically grounded approximation of the target sensor signals that captures coarse motion trends, but remains low-frequency and noisy due to limitations in visual estimation accuracy and video frame rates.

3.1.1 3D Pose Estimation. Given an RGB video sequence, our first step is to reconstruct the 3D trajectory of key object joints. We adopt a robust two-step lifting approach. First, we process each video frame with ViTPose++ [40], a powerful vision transformer-based model, to obtain accurate 2D keypoint coordinates for each object in the scene. Subsequently, the resulting 2D pose sequence is fed into VideoPose3D [22], a temporal convolutional network that leverages motion context to lift the 2D coordinates into a coherent 3D skeleton sequence. The output of this step is a sequence of 3D joint positions $P_{3D} \in \mathbb{R}^{T_v \times K \times 3}$, where T_v is the number of video frames and K is the number of keypoints.

3.1.2 Physics-Driven Kinematic Modeling. With the reconstructed 3D joint trajectories, we simulate the readings of a 6-axis IMU (3-axis accelerometer and 3-axis gyroscope) attached to a specific object segment. This simulation consists of three components: kinematic differentiation, coordinate system transformation, and MEMS noise modeling.

Kinematic Differentiation. This module computes the linear acceleration and angular velocity from the pose data. Let the 3D position of the distal joint of a segment (e.g., the wrist) at time t be denoted by the vector $\mathbf{p}(t)$. The global linear velocity $\mathbf{v}(t)$ and acceleration $\mathbf{a}_{\text{global}}(t)$ are obtained by taking the first and second time derivatives of the position, respectively:

$$\mathbf{v}(t) = \frac{d\mathbf{p}(t)}{dt}, \quad \mathbf{a}_{\text{global}}(t) = \frac{d^2\mathbf{p}(t)}{dt^2}. \quad (1)$$

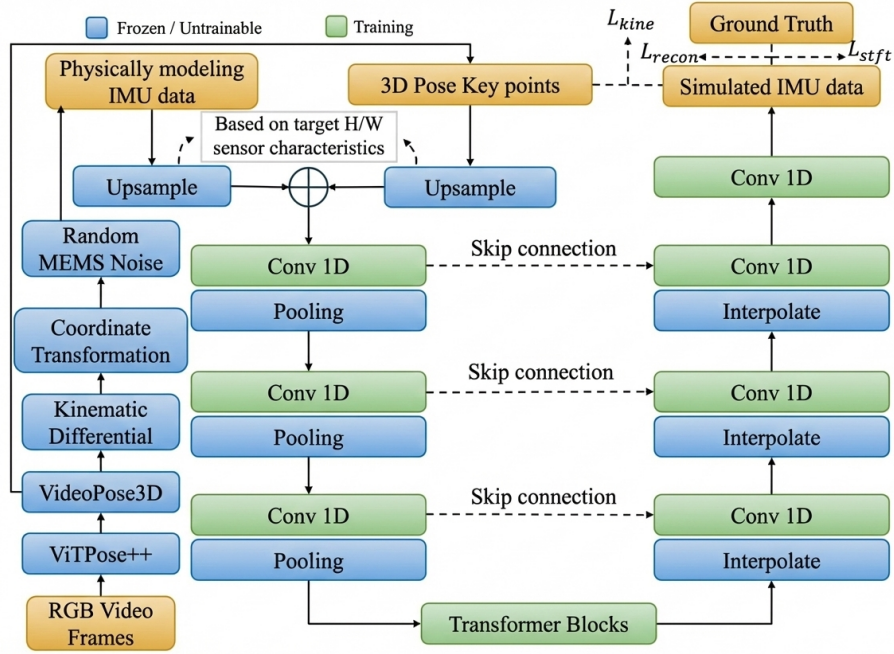


Fig. 2. Overview of the PrimelMU pipeline for synthesizing high-frequency IMU signals from RGB videos. **(Stage 1: Physics-based Simulation)** Each RGB frame is processed by ViTPose++ to estimate 2D keypoints, which are then lifted to 3D joint trajectories using VideoPose3D. The 3D poses are converted into low-frequency simulated IMU signals through kinematic differentiation, coordinate transformation, and MEMS noise modeling. **(Stage 2: Deep Generative Refinement)** A Hybrid U-Net with a transformer bottleneck takes the upsampled simulated IMU together with the upsampled pose sequence, refines them via convolutional–transformer blocks with skip connections, and generates realistic high-frequency IMU signals. Training is guided by a composite loss that combines time-domain reconstruction, frequency-domain STFT alignment, and kinematic consistency to enforce both fidelity and physical plausibility.

Directly differentiating discrete, noisy position data drastically amplifies noise. Therefore, we first apply a Savitzky-Golay filter [31] to the position sequence $\mathbf{p}(t)$ to obtain a smoothed trajectory before differentiation, which is implemented using finite differences.

The orientation of the object segment is more robustly defined by the vector connecting the proximal joint $\mathbf{p}_{\text{prox}}(t)$ to the distal joint $\mathbf{p}_{\text{dist}}(t)$. We use this vector to construct a time-varying local coordinate frame for the segment, represented by a sequence of orientation quaternions $q(t)$. The local angular velocity $\omega_{\text{local}}(t)$ is then derived from the temporal evolution of these quaternions. Given the relationship $\dot{q}(t) = \frac{1}{2} \begin{bmatrix} 0 \\ \omega_{\text{local}}(t) \end{bmatrix} \otimes q(t)$, where $\otimes$ denotes quaternion multiplication, the angular velocity can be computed as:

$$\begin{bmatrix} 0 \\ \omega_{\text{local}}(t) \end{bmatrix} = 2 \cdot \dot{q}(t) \otimes q(t)^{-1}, \quad (2)$$

where $q(t)^{-1}$ is the quaternion conjugate and $\dot{q}(t)$ is computed via finite differences.

Coordinate System Transformation. An IMU measures acceleration and angular velocity in its own local coordinate frame. The angular velocity $\omega_{\text{local}}(t)$ is already in this frame. However, the linear acceleration $\mathbf{a}_{\text{global}}(t)$ must be transformed. Using the orientation quaternion $q(t)$, we rotate the global acceleration vector into the local frame and subtract the effect of gravity, $\mathbf{g} = [0, 0, -9.81]^T \text{ m/s}^2$, which is also transformed into the local frame:

$$\mathbf{a}_{\text{local}}(t) = R(q(t))^{-1}(\mathbf{a}_{\text{global}}(t)) - R(q(t))^{-1}\mathbf{g}, \quad (3)$$

where $R(q(t))$ is the rotation matrix corresponding to quaternion $q(t)$. This step is crucial for simulating realistic accelerometer readings that include gravitational components.

MEMS Noise Modeling. Real IMU sensors are subject to various sources of noise. To enhance the realism of our simulated data, we model two primary noise types: a Gaussian white noise component and a random walk bias. The final simulated IMU signal $I_{\text{sim}}(t) = [\mathbf{a}_{\text{sim}}(t), \omega_{\text{sim}}(t)]$ is generated as:

$$\mathbf{a}_{\text{sim}}(t) = \mathbf{a}_{\text{local}}(t) + \mathbf{n}_a(t) + \mathbf{b}_a(t), \quad (4)$$

$$\omega_{\text{sim}}(t) = \omega_{\text{local}}(t) + \mathbf{n}_\omega(t) + \mathbf{b}_\omega(t), \quad (5)$$

where $\mathbf{n}(t)$ is sampled from a zero-mean Gaussian distribution whose standard deviation depends on the sensor's noise density, and $\mathbf{b}(t)$ is a bias term that evolves as a random walk. The output of this stage is a low-frequency simulated IMU signal, $I_{\text{sim}} \in \mathbb{R}^{T_o \times 6}$.

3.2 Stage 2: Hybrid U-Net Generative Refinement

The physics-guided simulation in Stage 1 produces an initial IMU estimate I_{sim} that is physically grounded but remains noisy and limited to the low temporal resolution of the source video. To bridge the gap between this coarse approximation and realistic high-frequency inertial measurements, the second stage employs a generative refinement network that learns to map low-quality simulated signals to high-quality IMU data, conditioned on the corresponding low-frequency pose sequence.

3.2.1 Network Architecture. We design a Hybrid U-Net architecture that combines the local feature extraction power of convolutions with the global context modeling of transformers. The network takes a low-frequency pose sequence $P_{\text{low}} \in \mathbb{R}^{T_l \times D_{\text{pose}}}$ and a random noise vector $N \in \mathbb{R}^{T_h \times D_{\text{imu}}}$ as input, where T_l and T_h are the lengths of the low- and high-frequency sequences, respectively.

The pose sequence P_{low} is first upsampled to the target length T_h via linear interpolation and then projected into a high-dimensional embedding space. This pose embedding is added to an embedding of the input noise N . The resulting sequence $X \in \mathbb{R}^{T_h \times D_{\text{model}}}$ is the input to the U-Net.

Encoder. The encoder consists of a sequence of 1D convolutional blocks with residual connections, each followed by max-pooling to progressively reduce temporal resolution. The feature maps produced before pooling are stored as skip connections, preserving fine-grained temporal details for later reconstruction.

Bottleneck. At the lowest temporal resolution, the encoded features are passed through a Transformer block. This bottleneck enables the model to capture long-range temporal dependencies and global motion patterns that are difficult to model using purely convolutional operations. Positional encodings are added prior to the Transformer to retain temporal ordering information.

Decoder. The decoder mirrors the encoder’s structure. At each stage, the feature map is upsampled using linear interpolation. The upsampled features are concatenated with the corresponding skip connection from the encoder path and passed through a 1D convolutional block. This process progressively refines the features while restoring the original temporal resolution. A final 1D convolutional layer projects the output features back to the dimension of the IMU data, yielding the generated high-frequency signal $I_{\text{gen}} \in \mathbb{R}^{T_h \times 6}$.

3.3 Composite Loss Design

The generative refinement network is trained end-to-end using a composite loss that jointly evaluates signal fidelity in the time and frequency domains while enforcing physical consistency with the input motion. The ground-truth supervision is provided by real high-frequency IMU measurements $I_{\text{gt}} \in \mathbb{R}^{T_h \times 6}$.

Reconstruction Loss ($\mathcal{L}_{\text{recon}}$). To enforce time-domain agreement between the generated signal and the ground truth, we apply an L1 reconstruction loss:

$$\mathcal{L}_{\text{recon}} = \|I_{\text{gen}} - I_{\text{gt}}\|_1. \quad (6)$$

This term penalizes deviations in signal amplitude and temporal structure.

STFT Loss ($\mathcal{L}_{\text{stft}}$). Accurately reproducing the frequency content of IMU signals is critical for realism. We therefore incorporate a multi-resolution Short-Time Fourier Transform (STFT) loss that aligns the spectral characteristics of the generated and real signals. The loss consists of a spectral convergence term and a log-magnitude term, computed across multiple Fast Fourier Transform (FFT) resolutions:

$$\mathcal{L}_{\text{stft}} = \sum_{r \in R} \left(\frac{\| |S_r(I_{\text{gen}})| - |S_r(I_{\text{gt}})| \|_F}{\| |S_r(I_{\text{gt}})| \|_F} + \|\log |S_r(I_{\text{gen}})| - \log |S_r(I_{\text{gt}})|\|_1 \right), \quad (7)$$

where $S_r(\cdot)$ denotes the STFT operator at resolution r , $|\cdot|$ is the magnitude, and $\|\cdot\|_F$ is the Frobenius norm.

Kinematic Consistency Loss ($\mathcal{L}_{\text{kine}}$). To ensure that the generated IMU signals remain physically consistent with the input motion, we introduce a kinematic consistency loss. Specifically, we extract the acceleration component of the generated signal, $I_{\text{gen}}^{\text{accel}} \in \mathbb{R}^{T_h \times 3}$, and apply a differentiable double integration operation to recover a high-frequency 3D trajectory $\hat{P}_{\text{high}}$. An initial position from the ground-truth low-frequency pose is used as the integration constant. The recovered trajectory is then downsampled to the original low temporal resolution, producing $\hat{P}_{\text{low}}$. The kinematic loss is defined as the Mean Squared Error (MSE) between this trajectory and the input low-frequency pose P_{low} that was an input to the network:

$$\mathcal{L}_{\text{kine}} = \|\text{Downsample}(\hat{P}_{\text{high}}) - P_{\text{low}}\|_2^2. \quad (8)$$

This term regularizes the generative process by enforcing consistency with the underlying kinematic motion.

Total Loss ($\mathcal{L}_{\text{total}}$). The overall training objective is a weighted sum of the three loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{stft}} \mathcal{L}_{\text{stft}} + \lambda_{\text{kine}} \mathcal{L}_{\text{kine}}, \quad (9)$$

where λ_{recon} , λ_{stft} , and λ_{kine} are hyper-parameters that balance the contribution of each term.

4 Experiments

4.1 Setup

4.1.1 Datasets. We conduct experiments on two publicly available multimodal benchmarks that provide both video and inertial sensor modalities for activity recognition: (i) **UTD-MHAD** [4] contains synchronized RGB videos, depth videos, skeleton joint positions, and 6-axis inertial signals (accelerometer and gyroscope) recorded from a Kinect camera and a single wearable inertial sensor. Eight subjects performed 27 different actions, including sports (e.g., basketball shoot, tennis swing), hand gestures (e.g., wave, clap, draw shapes), daily activities (e.g., sit-to-stand, knock on door), and exercises (e.g., squat, lunge, arm curl). Each subject repeated every action four times, yielding a total of 861 sequences. The inertial sensor was placed on the right wrist or thigh depending on the action, and all modalities were temporally synchronized and manually segmented, making UTD-MHAD a compact yet diverse benchmark for multimodal activity recognition. (ii) **MM-Fit** [33] contains synchronized multi-view RGB-D videos, skeleton poses, and wearable inertial signals (accelerometer, gyroscope, and magnetometer). Ten subjects performed ten types of resistance exercises, including squats, lunges, push-ups, sit-ups, curls, rows, presses, raises, and jumping jacks. Each exercise consisted of three sets of ten repetitions, resulting in 21 sessions, 616 sets, and 6160 annotated repetitions. All modalities were time-synchronized using a calibration jump and annotated with exercise types, set boundaries, and repetition counts, providing a challenging and diverse benchmark for multimodal activity recognition in workout scenarios.

4.1.2 Baseline. Because no prior state-of-the-art video-to-IMU methods publicly release their code, we construct a reproducible baseline by following the methodological descriptions provided in prior work, in particular IMUTube [13]. To the best of our knowledge, IMUTube represents a state-of-the-art approach for video-to-IMU synthesis. Due to the lack of an official implementation, we therefore re-implement its pipeline as faithfully as possible based on the details available in the paper. All references to the Baseline refer to this reconstructed IMUTube pipeline.

4.1.3 IMU Generation Quality Evaluation Metrics. We assess the fidelity of generated IMU signals by comparing them to ground-truth recordings using both error- and correlation-based measures. Error-based metrics (RMSE/MAE) quantify magnitude discrepancies but may overlook trend misalignment, while correlation-based metrics (R^2 and Pearson) ensure temporal and physical plausibility: (i) **Root Mean Square Error (RMSE)** emphasizes large deviations between generated and real signals, making it sensitive to spikes; small values indicate stable predictions, while large values suggest spiky or unstable outputs. (ii) **Mean Absolute Error (MAE)** captures the average absolute deviation from ground truth; lower values reflect better overall alignment, while higher values indicate consistent drift. (iii) **Coefficient of Determination (R^2)** measures the proportion of variance in the real signal explained by the generated one; values close to 1 indicate faithful reproduction of dynamics, values near 0 imply mean-level prediction, and negative values denote worse-than-mean performance. (iv) **Pearson Correlation** quantifies linear trend alignment between generated and ground-truth signals; values near +1 denote strong positive alignment, values near 0 indicate no relation, and values near -1 imply reversed, physically implausible trends.

4.1.4 Classification Performance Evaluation Metrics. We adopt two standard metrics for action recognition evaluation: (i) **Accuracy (Acc)** is the overall Top-1 classification accuracy, i.e., the fraction of correctly predicted activity labels across all samples. (ii) **Macro F1 (F1)** is the unweighted average of per-class F1 scores, giving equal importance to each class and thus providing robustness to class imbalance.

4.1.5 Classification Models. We evaluate the effectiveness of synthetic IMU data on five representative models spanning both classical machine learning and deep learning paradigms: (i) **Random Forest** [21] is an ensemble of decision trees that serves as a strong non-deep-learning baseline for IMU classification. (ii) **Support Vector Machine (SVM)** [35] is a kernel-based margin classifier that provides a competitive traditional baseline for time-series data. (iii) **DeepConvLSTM** [20] integrates convolutional layers for local feature extraction with LSTMs for temporal sequence modeling and is widely used in action recognition tasks. (iv) **DeepConvLSTM_Attention** [32] extends DeepConvLSTM by adding a self-attention mechanism to capture long-range temporal dependencies through adaptive weighting of time steps. (v) **Transformer** [37] is a fully attention-based architecture that models long-range dependencies without recurrence or convolutions and has recently gained popularity for sequential action recognition tasks. Together, these models form a diverse evaluation suite for assessing the utility of synthetic IMU data across classical and deep learning action recognition classifiers.

4.1.6 Configuration. For the IMU generation stage, the sampling frequency of the IMU data from the physics-based simulation is set to match the frame rate of the source video. Our U-Net model is trained for 100 epochs with a learning rate of 1×10^{-3} for both datasets. To evaluate the quality of our synthetic data, we use activity recognition as a downstream task to measure the performance gap between models trained on our generated data versus real IMU data. All neural classifiers in this downstream setting are trained for 50 epochs with a learning rate of 1×10^{-3} .

4.2 Comparison with the Baseline Method on IMU Generation Quality

Table 2. Comparison between PrimeIMU and a re-implemented baseline IMU generator on the UTD-MHAD and MM-Fit datasets. Results are reported per axis for accelerometer (X/Y/Z) and gyroscope (X/Y/Z) signals. IMU generation quality is evaluated using four complementary metrics: mean absolute error (MAE), root mean square error (RMSE), coefficient of determination (R^2), and Pearson correlation. For MAE and RMSE, lower values indicate better performance ($\downarrow$), while for R^2 and Pearson correlation, higher values indicate better agreement with real sensor measurements ($\uparrow$).

Dataset	Method	Metric	Accel X	Accel Y	Accel Z	Gyro X	Gyro Y	Gyro Z
UTD-MHAD	Baseline	MAE $\downarrow$	0.8395	7.6634	4.6674	1.3719	1.4953	2.1858
		RMSE $\downarrow$	1.0821	8.2838	5.3449	1.9902	2.2392	2.9975
		R^2 $\uparrow$	-670.6101	-65171.0273	-66641.8406	-1.1429	-1.3515	-2.9680
		Pearson $\uparrow$	-0.1797	0.1374	-0.1966	0.0143	0.0507	-0.1329
	PrimeIMU (Ours)	MAE $\downarrow$	0.0306	0.0176	0.0189	3.0549	3.1581	2.9964
		RMSE $\downarrow$	0.0399	0.0230	0.0262	4.2513	4.0785	3.8244
		R^2 $\uparrow$	0.8706	0.9785	0.9583	0.9842	0.8805	0.8852
		Pearson $\uparrow$	0.9886	0.9977	0.9916	0.9972	0.9957	0.9983
MM-Fit	Baseline	MAE $\downarrow$	2.1401	8.0887	6.4013	0.6339	0.7826	1.4799
		RMSE $\downarrow$	3.1713	9.7756	8.3973	0.9037	1.1181	2.2551
		R^2 $\uparrow$	-1754265773	-1540101166	-7826811739	-3045290257	-17280099211	-416054159758
		Pearson $\uparrow$	-0.0018	0.0016	-0.0032	-0.005	0.0005	-0.0047
	PrimeIMU (Ours)	MAE $\downarrow$	0.2808	0.5844	0.2074	0.0691	0.0341	0.0495
		RMSE $\downarrow$	0.3656	0.6386	0.2764	0.0890	0.0442	0.0655
		R^2 $\uparrow$	-0.1784	-6.4791	0.3422	-0.1988	0.5433	-1.8088
		Pearson $\uparrow$	0.8963	0.9275	0.9333	0.9455	0.9526	0.9200

We compare PrimeIMU with a re-implemented kinematics-only baseline on two multimodal benchmarks, UTD-MHAD and MM-Fit, to evaluate IMU generation quality under both in-domain and cross-domain conditions. Results are reported per axis for accelerometer (X/Y/Z) and gyroscope (X/Y/Z) signals using four complementary metrics: mean absolute error (MAE), root mean square error (RMSE), coefficient of determination (R^2), and Pearson correlation

(Table 2). Together, these metrics jointly assess amplitude accuracy, variance alignment, and temporal fidelity with respect to real sensor measurements.

Across both datasets, PrimeIMU consistently outperforms the baseline by large margins on all correlation-based metrics, while simultaneously reducing absolute errors. In contrast, the baseline frequently exhibits near-zero or negative Pearson correlations together with extremely negative R^2 values, indicating a failure to capture meaningful temporal structure. PrimeIMU, by comparison, produces signals that closely align with real IMU trajectories over time and explain a substantial portion of signal variance across nearly all axes.

On UTD-MHAD, PrimeIMU achieves near-ceiling temporal fidelity across all six sensor axes. Pearson correlations improve from baseline values ranging between -0.20 and 0.14 to above 0.99 for both accelerometer and gyroscope channels. Correspondingly, R^2 increases from extremely negative values (down to -6.6×10^4 on accelerometer axes) to consistently positive values between 0.87 and 0.98 , indicating strong variance alignment with real sensor signals. Absolute errors are also substantially reduced: accelerometer MAE decreases from 0.84 – 7.66 to below 0.03 , and RMSE from 1.08 – 8.28 to below 0.04 , representing more than an order-of-magnitude reduction on most axes.

Results on MM-Fit further highlight the robustness of PrimeIMU under stronger domain shift. MM-Fit involves higher variability in exercises, sensor placements, and motion intensities, which more closely reflect real-world deployment conditions. Under this setting, the baseline effectively loses temporal structure, yielding Pearson correlations near zero (within ± 0.005) and extremely negative R^2 values (as low as -4.2×10^{11}). In contrast, PrimeIMU maintains strong temporal alignment across all axes, with Pearson correlations consistently between 0.90 and 0.95 . Absolute errors are also sharply reduced: accelerometer MAE drops from 2.14 – 8.09 to 0.21 – 0.58 , and RMSE from 3.17 – 9.78 to 0.28 – 0.64 .

For accelerometer signals, these improvements indicate that PrimeIMU recovers both realistic temporal dynamics and accurate amplitude scaling. Across both datasets, accelerometer MAE and RMSE are reduced by factors ranging from approximately $10\times$ to over $30\times$, while Pearson correlations improve from near-zero or negative values to above 0.90 . This suggests that translational motion patterns, which are tightly constrained by underlying kinematics and physics, are effectively captured by the combined physics-guided initialization and learned refinement stages.

For gyroscope signals, PrimeIMU likewise achieves strong temporal fidelity. On UTD-MHAD, Pearson correlations exceed 0.99 across all gyroscope axes, with R^2 values reaching up to 0.98 . While absolute gyroscope errors (MAE/RMSE) do not always decrease monotonically across all axes, the simultaneous presence of near-ceiling correlation and high variance explanation indicates that remaining discrepancies primarily reflect scale or bias differences rather than errors in temporal dynamics. Such behavior is consistent with device-specific angular rate scaling and mounting-dependent effects, and can typically be addressed through lightweight calibration without altering downstream recognition performance. On MM-Fit, PrimeIMU dominates the baseline on both error-based and correlation-based metrics, with gyroscope RMSE reduced by up to an order of magnitude (e.g., Gyro-X $0.90 \rightarrow 0.09$) and Pearson correlations improving from near zero to approximately 0.95 .

Overall, the quantitative results demonstrate that PrimeIMU substantially outperforms a kinematics-only baseline in generating realistic and sensor-faithful IMU signals. The most pronounced gains occur in correlation-based measures, where PrimeIMU consistently recovers correct temporal dynamics and variance structure that the baseline fails to model. Importantly, these improvements persist across datasets with substantial differences in sensing hardware, activity distributions, and execution styles, confirming that integrating physics-guided simulation with learned high-frequency refinement yields inertial signals that are not only numerically accurate, but also dynamically meaningful for downstream applications.

4.3 Classification Performance using Generated IMU Data: Baseline vs. Ours

Table 3. Classification results on the UTD-MHAD dataset comparing a re-implemented baseline IMU generator and PrimeIMU across four training–testing configurations: GT→GT (training and testing on real IMU data), GEN→GEN (training and testing on generated IMU data), GT→GEN (training on real IMU and testing on generated IMU), and GEN→GT (training on generated IMU and testing on real IMU). Results are reported for five classifiers using Accuracy (Acc) and macro-F1. Higher values indicate better performance (↑).

Model	Train	Test	Baseline		PrimeIMU (Ours)	
			Acc ↑	F1 ↑	Acc ↑	F1 ↑
Random Forest	GT	GT	0.8023	0.7962	0.8023	0.7962
	GEN	GEN	0.6734	0.6737	0.8166	0.8044
	GT	GEN	0.0201	0.0041	0.7880	0.7813
	GEN	GT	0.0516	0.0223	0.7937	0.7847
SVM	GT	GT	0.7106	0.6908	0.7106	0.6908
	GEN	GEN	0.4785	0.4711	0.7106	0.6882
	GT	GEN	0.0602	0.0042	0.7221	0.7052
	GEN	GT	0.0287	0.0021	0.6991	0.6747
DeepConvLSTM	GT	GT	0.7564	0.7612	0.7564	0.7612
	GEN	GEN	0.4871	0.4793	0.7421	0.7303
	GT	GEN	0.0372	0.0057	0.7564	0.7537
	GEN	GT	0.0201	0.0124	0.7278	0.7119
DeepConvLSTM_Attention	GT	GT	0.8940	0.8907	0.8940	0.8907
	GEN	GEN	0.7736	0.7744	0.9083	0.9033
	GT	GEN	0.0315	0.0024	0.8854	0.8750
	GEN	GT	0.0401	0.0168	0.8863	0.8813
Transformer	GT	GT	0.8625	0.8624	0.8625	0.8624
	GEN	GEN	0.7679	0.7717	0.8625	0.8563
	GT	GEN	0.0315	0.0023	0.8453	0.8396
	GEN	GT	0.0659	0.0236	0.8453	0.8375

We evaluate whether generated IMU signals can effectively support downstream activity recognition and, critically, whether they reduce the distribution gap between synthetic and real sensor data. Following prior work, we consider four training–testing configurations: GT→GT (training and testing on real IMU), GEN→GEN (training and testing on generated IMU), GT→GEN (training on real IMU and testing on generated IMU), and GEN→GT (training on generated IMU and testing on real IMU). Results on UTD-MHAD and MM-Fit are summarized in Tables 3 and 4 across five representative classifiers, covering both classical models and deep temporal architectures.

Across both datasets and all classifiers, PrimeIMU consistently outperforms the kinematics-only baseline in every configuration involving synthetic data. This trend holds for Random Forest and SVM as well as for DeepConvLSTM variants and Transformer-based models, indicating that the gains are not tied to a specific classifier architecture but instead arise from improved signal fidelity at the representation level. In particular, PrimeIMU yields strong and symmetric performance in both GT→GEN and GEN→GT settings, suggesting a substantial reduction in the synthetic–real domain gap.

On UTD-MHAD, PrimeIMU enables synthetic-only training to closely approach the real-data reference. In the deployment-relevant GEN→GT setting, the baseline collapses across all models, yielding near-random accuracy and macro-F1 (e.g., Random Forest 0.0516/0.0223, SVM 0.0287/0.0021, Transformer 0.0659/0.0236). In contrast, PrimeIMU raises GEN→GT performance to levels comparable to GT→GT. For example, the Transformer achieves 0.8453/0.8375 under GEN→GT compared to 0.8625/0.8624 under GT→GT, while DeepConvLSTM_Attention reaches 0.8863/0.8813

Table 4. Classification results on the MM-Fit dataset comparing a re-implemented baseline IMU generator and PrimeIMU across four training-testing configurations: GT→GT (training and testing on real IMU data), GEN→GEN (training and testing on generated IMU data), GT→GEN (training on real IMU and testing on generated IMU), and GEN→GT (training on generated IMU and testing on real IMU). Results are reported for five classifiers using Accuracy (Acc) and macro-F1. Higher values indicate better performance (↑).

Model	Train	Test	Baseline		PrimeIMU (Ours)	
			Acc ↑	F1 ↑	Acc ↑	F1 ↑
Random Forest	GT	GT	0.4034	0.3833	0.4034	0.3833
	GEN	GEN	0.2149	0.1613	0.3556	0.3345
	GT	GEN	0.1596	0.1287	0.3499	0.3073
	GEN	GT	0.1664	0.1446	0.3614	0.3342
SVM	GT	GT	0.3572	0.3106	0.3572	0.3106
	GEN	GEN	0.1166	0.0729	0.3512	0.3522
	GT	GEN	0.1088	0.1483	0.3103	0.3013
	GEN	GT	0.1357	0.1138	0.3492	0.3086
DeepConvLSTM	GT	GT	0.3686	0.3488	0.3686	0.3488
	GEN	GEN	0.2461	0.1890	0.3429	0.3063
	GT	GEN	0.1437	0.1140	0.3218	0.2777
	GEN	GT	0.0988	0.0770	0.3322	0.3031
DeepConvLSTM_Attention	GT	GT	0.3541	0.3374	0.3541	0.3374
	GEN	GEN	0.2851	0.2176	0.3446	0.3044
	GT	GEN	0.1587	0.1485	0.2993	0.2667
	GEN	GT	0.1624	0.1267	0.3144	0.2957
Transformer	GT	GT	0.3787	0.3631	0.3787	0.3631
	GEN	GEN	0.2871	0.2106	0.3442	0.2926
	GT	GEN	0.0743	0.0759	0.3086	0.2719
	GEN	GT	0.1376	0.1143	0.3255	0.3030

versus 0.8940/0.8907. The remaining gap is consistently small, typically within a few percentage points, indicating that PrimeIMU preserves most of the discriminative structure required for recognition even when no real IMU data are used during training.

The reverse direction, GT→GEN, provides a complementary assessment of realism. If synthetic signals resemble real IMU in task-relevant ways, models trained on real data should retain performance when evaluated on synthetic inputs. This is not the case for the baseline, where GT→GEN accuracy is near zero across all classifiers. PrimeIMU, however, yields GT→GEN results close to GT→GT, such as 0.8453/0.8396 for the Transformer and 0.7880/0.7813 for Random Forest. Together with the GEN→GT results, this bidirectional transfer indicates that PrimeIMU substantially reduces the distribution mismatch between synthetic and real IMU domains in a manner that directly benefits downstream recognition.

Within-domain learnability is reflected by the GEN→GEN setting. PrimeIMU improves GEN→GEN performance over the baseline across all models, often by large margins (e.g., DeepConvLSTM 0.4871/0.4793 → 0.7421/0.7303, Random Forest 0.6734/0.6737 → 0.8166/0.8044). Notably, DeepConvLSTM_Attention trained and tested on PrimeIMU signals slightly exceeds its GT→GT reference. This behavior likely reflects increased temporal regularity in the generated signals, which can simplify optimization in a fully synthetic setting. Importantly, this effect is confined to within-domain evaluation and does not suggest that synthetic IMU surpasses real sensor data for real-world deployment.

On MM-Fit, absolute accuracies are lower for all methods, consistent with the dataset’s greater variability in exercises, devices, and sensor placements. This setting more closely reflects practical deployment scenarios, where heterogeneity is

unavoidable. Despite this increased difficulty, PrimeIMU consistently outperforms the baseline across all configurations and classifiers. In the GEN→GT setting, PrimeIMU improves accuracy by approximately 15–23 points across models (e.g., Transformer 0.1376 → 0.3255, DeepConvLSTM 0.0988 → 0.3322, SVM 0.1357 → 0.3492), with corresponding macro-F1 gains of similar magnitude. In the GT→GEN setting, PrimeIMU also yields substantial improvements (e.g., Transformer 0.0743/0.0759 → 0.3086/0.2719), indicating that real-trained classifiers remain effective when evaluated on PrimeIMU-generated signals. GEN→GEN results further confirm that the synthetic IMU distribution produced by PrimeIMU is internally coherent and learnable, whereas the baseline remains far from competitive.

Overall, these results demonstrate that PrimeIMU-generated IMU signals are not only learnable within the synthetic domain, but also transferable across synthetic and real domains in both directions. The improvements are most pronounced in the deployment-relevant synthetic-only training scenario, where PrimeIMU enables classifiers to generalize effectively to real sensor data without access to real IMU during training. This substantially reduces reliance on large-scale labeled IMU collection and supports the practical use of PrimeIMU for scalable IMU-based activity recognition.

4.4 Data Augmentation

We study whether PrimeIMU can serve as an effective augmentation source when combined with limited real IMU data. Table 5 reports action recognition results on UTD-MHAD when models are trained on real IMU alone (GT) versus a mixture of real and synthetic data (GT+GEN), and evaluated on real IMU.

Table 5. Data augmentation results on the UTD-MHAD dataset. Models are trained either on real IMU data only (GT) or on a combination of real and PrimeIMU-generated synthetic data (GT+GEN), and evaluated on real IMU signals. We report classification accuracy and macro F1 across classical and deep sequential models, highlighting the impact of synthetic IMU augmentation on recognition performance.

Model	Training	Testing	Accuracy ↑	F1 Score (Macro) ↑
RandomForest	GT	GT	0.8023	0.7962
	GT+GEN	GT	0.8381	0.8266
SVM	GT	GT	0.7106	0.6908
	GT+GEN	GT	0.7550	0.7413
DeepConvLSTM	GT	GT	0.7564	0.7612
	GT+GEN	GT	0.8840	0.8743
DeepConvLSTM_Attention	GT	GT	0.8940	0.8907
	GT+GEN	GT	0.9284	0.9247
Transformer	GT	GT	0.8625	0.8624
	GT+GEN	GT	0.8467	0.8483

Across four out of five models, augmenting real training data with PrimeIMU-generated signals consistently improves performance. The gains range from modest improvements for classical classifiers to substantial boosts for deep temporal models, indicating that synthetic IMU signals provide complementary information beyond what is available in the original dataset.

For classical models, augmentation yields stable and meaningful improvements. RandomForest accuracy increases from 0.8023 to 0.8381 (+3.6 points), with macro F1 rising from 0.7962 to 0.8266. SVM shows a similar pattern, improving from 0.7106 to 0.7550 accuracy (+4.4 points) and from 0.6908 to 0.7413 macro F1. These gains suggest that PrimeIMU

expands the effective coverage of the feature space, even for shallow models that rely on handcrafted or aggregated temporal features.

The impact is more pronounced for deep recurrent architectures. DeepConvLSTM benefits substantially from augmentation, with accuracy improving from 0.7564 to 0.8840 (+12.8 points) and macro F1 from 0.7612 to 0.8743. DeepConvLSTM_Attention also sees consistent gains, increasing accuracy from 0.8940 to 0.9284 (+3.4 points) and macro F1 from 0.8907 to 0.9247. These results indicate that recurrent temporal models are particularly effective at exploiting the additional fine-grained motion patterns introduced by synthetic IMUs, translating augmented temporal diversity into better generalization on real data.

In contrast, the Transformer shows a small decrease when synthetic data is added ($0.8625 \rightarrow 0.8467$ accuracy). Given its already strong baseline performance, this result suggests that naive augmentation may introduce distributional variability that is not uniformly beneficial for globally attentive architectures. Importantly, this effect is limited to a single model and does not negate the overall trend observed across the remaining classifiers.

Taken together, these findings demonstrate that PrimeIMU is not only suitable for synthetic-only training, but also serves as a practical augmentation mechanism when some real IMU data are available. Augmentation consistently improves or maintains performance for most models, with particularly large gains for recurrent temporal architectures. This suggests that PrimeIMU-generated signals enrich the training distribution in a way that complements real sensor measurements, helping models learn more robust motion representations under limited data regimes.

4.5 Evaluation of Unseen Activity Generation Quality

We further evaluate the generalization ability of PrimeIMU under activity-level distribution shift, focusing on whether realistic IMU signals can be synthesized for activities that are entirely absent during training. This setting reflects a practical deployment scenario, where exhaustive coverage of all activities is infeasible and the model must extrapolate to new motion patterns. To simulate this condition, we progressively remove increasing numbers of activity classes from the training set and evaluate generation quality on the held-out classes. Specifically, we consider four configurations where 1, 3, 7, or 13 activity classes are excluded during training. Performance is assessed using MAE, RMSE, coefficient of determination (R^2), and Pearson correlation across all accelerometer and gyroscope axes. Results are summarized in Table 6.

Generation quality degrades gradually as more activity classes are withheld, reflecting the increasing difficulty of extrapolating to unseen motion patterns. Importantly, this degradation is not catastrophic. Even in the most challenging setting, where 13 out of 27 activities are removed during training, PrimeIMU preserves strong temporal alignment with real sensor signals, as indicated by consistently high Pearson correlations, typically in the range of 0.95–0.98 across both accelerometer and gyroscope channels. High Pearson correlation in this context is particularly important, as it indicates that the temporal evolution and phase structure of motion dynamics are preserved under activity-level distribution shift, even when absolute signal magnitudes vary.

Accelerometer signals exhibit strong robustness to unseen activities. When only a small number of classes are removed (1 or 3), MAE and RMSE remain very low (approximately 0.02–0.04), and Pearson correlations exceed 0.92 across all axes. Even under the most severe removal of 13 activity classes, accelerometer correlations remain high (0.96–0.98), with MAE below 0.05 and largely positive R^2 values. This robustness suggests that translational motion patterns, which are closely tied to underlying kinematic trajectories and gravity constraints, generalize well across activities and are effectively captured by PrimeIMU’s physics-guided simulation and refinement pipeline.

Table 6. Unseen activity generation performance of PrimeIMU under increasing activity-level distribution shift. We progressively remove different numbers of activity classes during training (1, 3, 7, and 13 classes) and evaluate IMU generation quality on the held-out activities. Results are reported per axis for accelerometer and gyroscope signals using MAE, RMSE, coefficient of determination (R^2), and Pearson correlation, capturing both magnitude accuracy and temporal alignment.

Configuration	Dimension	MAE↓	RMSE↓	R^2 ↑	Pearson↑
w/o 1 class	Accel-X	0.0278	0.0318	-4.5762	0.8223
	Accel-Y	0.0319	0.0347	0.5526	0.9680
	Accel-Z	0.0178	0.0351	-0.9815	0.7919
	Gyro-X	3.1941	4.1490	0.8473	0.9549
	Gyro-Y	2.1869	2.7867	0.4120	0.9200
	Gyro-Z	3.3438	4.5424	0.5921	0.9442
w/o 3 classes	Accel-X	0.0200	0.0249	0.3460	0.9206
	Accel-Y	0.0152	0.0215	0.7880	0.9614
	Accel-Z	0.0206	0.0253	0.3325	0.9298
	Gyro-X	6.3002	6.7170	0.5306	0.9809
	Gyro-Y	6.1725	6.8002	-3.6676	0.8741
	Gyro-Z	3.6351	4.5474	0.2053	0.9547
w/o 7 classes	Accel-X	0.0470	0.0639	-0.6917	0.8887
	Accel-Y	0.0321	0.0457	0.6769	0.9643
	Accel-Z	0.0328	0.0440	0.3705	0.9258
	Gyro-X	5.5519	7.3369	0.4505	0.9772
	Gyro-Y	4.5878	5.9229	-0.7792	0.9254
	Gyro-Z	9.1852	10.4794	-0.7322	0.9777
w/o 13 classes	Accel-X	0.0489	0.0651	0.2695	0.9594
	Accel-Y	0.0501	0.0647	0.5779	0.9779
	Accel-Z	0.0466	0.0592	0.1732	0.9558
	Gyro-X	9.3833	11.8290	-0.8209	0.9755
	Gyro-Y	7.5312	9.5950	-0.6213	0.9545
	Gyro-Z	9.6361	11.5002	-1.4849	0.9751

Gyroscope signals show greater sensitivity to unseen activities, particularly as the number of removed classes increases. With only one class excluded, gyroscope channels already achieve strong agreement with real data, with Pearson correlations above 0.92 and positive R^2 across all axes. As more activities are removed, MAE and RMSE increase noticeably, and R^2 becomes negative on several axes (e.g., Gyro-Z under 7- and 13-class removal). These negative R^2 values primarily reflect scale and variance mismatch rather than loss of temporal correspondence, as temporal correlations remain consistently high (0.92–0.98). This behavior is consistent with the fact that angular velocity magnitudes are strongly coupled to activity-specific execution speed, rotation range, and articulation style, and are therefore less directly constrained by kinematic trajectories alone.

Overall, the unseen-activity evaluation highlights three key findings. First, PrimeIMU maintains strong temporal fidelity for both accelerometer and gyroscope signals even when synthesizing IMU data for activities that are entirely unseen during training. Second, accelerometer magnitudes generalize robustly across activities, while gyroscope channels exhibit larger amplitude deviations under heavy activity removal, suggesting that lightweight calibration may further improve fidelity in extreme cases. Third, the absence of catastrophic failure under large activity-level distribution shifts indicates that PrimeIMU learns transferable inertial primitives and motion dynamics rather than memorizing activity-specific signal templates. This property is essential for scalable IMU synthesis in real-world settings, where exhaustive activity coverage during training is impractical.

Table 7. Cross-dataset IMU generation quality when transferring from UTD-MHAD to MM-Fit. Models are pretrained on UTD-MHAD and fine-tuned with different proportions of MM-Fit data (10%, 20%, 30%, 40%). We report signal-level metrics across six sensor dimensions: mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and Pearson correlation. These generated signals serve as the basis for the classification experiments in Table 8.

Dimension	Data Proportion	MAE ↓	RMSE ↓	R^2 ↑	Pearson ↑
Accel-X	10%	<u>0.1448</u>	<u>0.1934</u>	0.4315	0.9261
	20%	0.1515	0.1993	0.7276	0.9571
	30%	0.1477	0.1946	0.4309	0.9247
	40%	0.1294	0.1797	<u>0.6439</u>	<u>0.9416</u>
Accel-Y	10%	0.4493	0.4911	-5.5504	0.8799
	20%	0.5704	0.7809	<u>-0.0863</u>	0.6772
	30%	<u>0.2259</u>	<u>0.2699</u>	0.2087	0.9511
	40%	0.1935	0.2626	-0.3392	<u>0.9116</u>
Accel-Z	10%	0.2059	0.2499	-0.0960	0.9444
	20%	0.1146	0.1451	<u>0.5209</u>	0.9743
	30%	0.1904	0.2321	0.3915	0.9528
	40%	<u>0.1191</u>	<u>0.1619</u>	0.7264	<u>0.9533</u>
Gyro-X	10%	<u>0.0500</u>	<u>0.0634</u>	<u>0.2247</u>	<u>0.9760</u>
	20%	0.0342	0.0418	0.7645	0.9827
	30%	0.0558	0.0679	-0.6523	0.9575
	40%	0.0667	0.0783	0.0155	0.9747
Gyro-Y	10%	0.0801	0.0863	-9.4137	0.9598
	20%	0.0266	0.0318	<u>0.2922</u>	<u>0.9738</u>
	30%	<u>0.0442</u>	<u>0.0506</u>	-0.3174	0.9728
	40%	0.0288	0.0352	0.5606	0.9787
Gyro-Z	10%	0.0538	0.0675	-1.5817	0.9440
	20%	0.0446	0.0513	0.6121	0.9746
	30%	<u>0.0531</u>	<u>0.0585</u>	<u>0.1843</u>	<u>0.9622</u>
	40%	0.0591	0.0680	-5.4008	0.9633

4.6 Cross-Dataset Transfer Learning for IMU Generation and Classification

A central requirement for practical IMU synthesis is robust cross-dataset generalization: models trained on one dataset should adapt to new domains with different devices, sampling rates, sensor characteristics, and activity styles using minimal target-domain data. We evaluate this setting by pretraining PrimeIMU on UTD-MHAD and transferring it to MM-Fit with 10%, 20%, 30%, and 40% of target-domain data. Signal-level IMU generation quality is reported in Table 7, while downstream action recognition performance is summarized in Table 8. Together, these experiments assess whether PrimeIMU preserves both physically meaningful inertial structure and task-relevant discriminative information under substantial domain shift.

4.6.1 Signal-Level Cross-Dataset Adaptation. We first analyze cross-dataset generalization at the signal level using Table 7, which reports IMU generation quality when PrimeIMU is pretrained on UTD-MHAD and fine-tuned on MM-Fit with increasing proportions of target-domain data. This analysis isolates whether inertial structure learned from one dataset can be transferred to another dataset with distinct sensor hardware, sampling frequencies, exercise protocols, and subject execution styles.

With only 10% of MM-Fit data, PrimeIMU already demonstrates strong temporal alignment with real IMU signals. Pearson correlations exceed 0.92 on most axes, including Accel-X (0.926), Accel-Z (0.944), Gyro-X (0.976), and Gyro-Y (0.960). This indicates that the temporal evolution and relative phase structure of inertial motion are largely dataset-invariant and can be transferred from UTD-MHAD to MM-Fit with minimal supervision. At the same time, variance-based metrics such as R^2 remain low or negative on several axes (e.g., Accel-Y -5.55 , Gyro-Y -9.41), reflecting residual discrepancies in signal magnitude and sensor-specific scaling. The coexistence of high Pearson correlation and low R^2 suggests that PrimeIMU captures correct motion dynamics early, while absolute amplitude alignment to the target device is not yet fully calibrated.

Increasing the fine-tuning ratio to 20% yields clear and consistent improvements across both accelerometer and gyroscope channels. For example, Accel-Z MAE decreases from 0.206 to 0.115 and RMSE from 0.250 to 0.145, while R^2 becomes positive (0.52) and Pearson increases to 0.974. Similarly, Gyro-X improves to MAE 0.034 with $R^2 = 0.76$ and Pearson 0.983. These results demonstrate that a modest amount of target-domain data is sufficient to substantially correct magnitude-related discrepancies on top of already well-aligned temporal structure. The improvements are axis-dependent: accelerometer channels tend to stabilize earlier, whereas gyroscope channels—more sensitive to angular velocity scaling, sensor placement, and exercise execution—require additional supervision to achieve reliable variance alignment.

Beyond 20% fine-tuning, further increases in target-domain data do not lead to consistent gains in signal quality. Pearson correlations remain stably high across all axes, typically in the range of 0.95–0.98, indicating that temporal motion dynamics are preserved throughout. In contrast, variance-based metrics exhibit non-monotonic behavior, particularly for gyroscope channels. For instance, Gyro-Z R^2 decreases from 0.18 at 30% to -5.40 at 40%, despite nearly unchanged Pearson correlation. We attribute this mild non-monotonicity to the interaction between limited target-domain scale diversity and device-specific normalization, rather than to changes in the underlying temporal modeling. Importantly, the stability of Pearson correlation across all fine-tuning ratios confirms that PrimeIMU maintains realistic motion dynamics even when exact amplitude alignment varies.

Overall, Table 7 reveals a consistent pattern: cross-dataset transfer of temporal inertial structure is highly data-efficient, with strong temporal fidelity achieved using as little as 10–20% target-domain data. Magnitude calibration improves rapidly at early stages but does not monotonically benefit from additional fine-tuning. This separation between temporal fidelity and amplitude alignment is particularly relevant for downstream recognition tasks, which depend more strongly on relative motion dynamics and inter-axis coordination than on exact signal scaling.

4.6.2 Downstream Classification under Cross-Dataset Transfer. We next examine whether the signal-level cross-dataset generalization observed in Table 7 translates into downstream utility for action recognition. Table 8 reports classification performance on MM-Fit under four training–testing configurations using PrimeIMU-generated signals at different fine-tuning ratios.

The most practically relevant setting is GEN→GT, where classifiers are trained entirely on synthetic IMU signals and evaluated on real MM-Fit data. Across all model families, performance under this setting consistently approaches the real-data reference (GT→GT), even with limited target-domain fine-tuning. For example, at 20% fine-tuning, the Transformer achieves an accuracy/F1 of 0.331/0.306 under GEN→GT, compared to 0.379/0.351 under GT→GT. DeepConvLSTM exhibits a similar trend, reaching 0.334/0.312 versus 0.369/0.349. Classical models follow the same pattern: SVM maintains GEN→GT accuracy around 0.35 across all fine-tuning ratios, only marginally below its GT→GT

Table 8. Action recognition on MM-Fit using synthetic IMU signals generated in Table 7. We evaluate four train/test configurations: (1) GT→GT, training and testing on ground-truth IMU; (2) GEN→GEN, training and testing on PrimeIMU-generated IMU; (3) GT→GEN, trained on real and tested on synthetic; and (4) GEN→GT, trained on synthetic and tested on real. Results are reported under different fine-tuning proportions of MM-Fit (10%, 20%, 30%, 40%).

Model	Train	Test	10%		20%		30%		40%	
			Acc ↑	F1 ↑	Acc ↑	F1 ↑	Acc ↑	F1 ↑	Acc ↑	F1 ↑
Random Forest	GT	GT	0.4034	0.3833	0.4034	0.3833	0.4034	0.3833	0.4034	0.3833
	GEN	GEN	0.3490	0.3241	0.3508	0.3278	0.3481	0.3268	0.3495	0.3255
	GT	GEN	0.3582	0.3176	0.3569	0.3182	0.3591	0.3222	0.3675	0.3289
	GEN	GT	0.3663	0.3404	0.3625	0.3382	0.3681	0.3455	0.3711	0.3477
SVM	GT	GT	0.3572	0.3106	0.3572	0.3106	0.3572	0.3106	0.3572	0.3106
	GEN	GEN	0.3508	0.3518	0.3512	0.3531	0.3530	0.3533	0.3516	0.3525
	GT	GEN	0.3112	0.3020	0.3081	0.2989	0.3081	0.2990	0.3055	0.2917
	GEN	GT	0.3505	0.3084	0.3503	0.3086	0.3524	0.3116	0.3524	0.3110
DeepConvLSTM	GT	GT	0.3686	0.3488	0.3686	0.3488	0.3684	0.3423	0.3684	0.3423
	GEN	GEN	0.3556	0.3009	0.3424	0.3134	0.3530	0.3003	0.3235	0.2805
	GT	GEN	0.3174	0.2708	0.3213	0.2792	0.3363	0.2866	0.3275	0.2800
	GEN	GT	0.3427	0.3177	0.3337	0.3123	0.3604	0.3328	0.3461	0.3192
DeepConvLSTM_Attention	GT	GT	0.3541	0.3374	0.3642	0.3440	0.3568	0.3405	0.3568	0.3405
	GEN	GEN	0.3481	0.3103	0.3231	0.2775	0.3429	0.2834	0.3371	0.3023
	GT	GEN	0.3160	0.2724	0.3156	0.2736	0.3182	0.2733	0.3191	0.2777
	GEN	GT	0.3385	0.3151	0.3274	0.3058	0.3335	0.3127	0.3293	0.3052
Transformer	GT	GT	0.3787	0.3631	0.3787	0.3507	0.3740	0.3507	0.3740	0.3507
	GEN	GEN	0.3240	0.2694	0.3490	0.3089	0.3389	0.2849	0.3411	0.2998
	GT	GEN	0.3310	0.2813	0.3279	0.2716	0.3349	0.2847	0.3292	0.2806
	GEN	GT	0.3314	0.3014	0.3312	0.3059	0.3448	0.3168	0.3280	0.3027

performance of 0.357. These results indicate that PrimeIMU preserves the discriminative temporal patterns required for real-world deployment, even when no real IMU data are used during classifier training.

The GEN→GEN setting provides complementary evidence of internal consistency within the synthetic domain. Models trained and tested entirely on PrimeIMU-generated signals achieve stable and competitive performance across fine-tuning ratios. Notably, SVM reaches 0.353/0.353 at 30% fine-tuning, approaching its GT→GT baseline. Similar trends are observed for Random Forest and deep sequence models, demonstrating that the synthetic IMU distribution is not only realistic relative to real sensors but also coherent and learnable in its own right.

The GT→GEN configuration evaluates whether synthetic signals are perceived as realistic by models trained on real IMU data. Performance remains stable across fine-tuning ratios, with moderate and controlled gaps relative to GT→GT. For instance, the Transformer achieves 0.335/0.285 at 30% fine-tuning, compared to 0.379/0.363 under GT→GT. Unlike prior kinematics-only synthesis approaches, where GT-trained models often degrade sharply on synthetic inputs, PrimeIMU-generated signals remain compatible with real-trained classifiers. This behavior aligns with the signal-level observation that temporal motion dynamics are faithfully preserved across domains.

Across all classifiers, classification performance improves rapidly between 10% and 20% fine-tuning and then stabilizes, mirroring the saturation behavior observed in signal-level Pearson correlation and temporal metrics. This consistency suggests that downstream recognition models primarily rely on relative temporal structure and inter-axis coordination, which are already well-aligned at low fine-tuning ratios. Consequently, exact amplitude calibration, while beneficial for signal fidelity, plays a secondary role in enabling cross-dataset transfer for IMU-based activity recognition.

Taken together, the results demonstrate that PrimeIMU enables effective cross-dataset transfer at both the signal and task levels. With limited target-domain data, the framework captures dataset-invariant inertial dynamics that support robust action recognition across heterogeneous sensor configurations, substantially reducing the need for large-scale recollection of labeled IMU data when deploying recognition systems in new environments.

4.7 Ablation Analysis

We conduct an ablation study on UTD-MHAD to quantify the contribution of PrimeIMU’s core design components. Specifically, we ablate (i) input sources, including low-frequency pose guidance and physics-inspired simulated IMU initialization, and (ii) objective terms, including time-domain reconstruction loss $\mathcal{L}_{\text{recon}}$, multi-resolution spectral alignment loss $\mathcal{L}_{\text{stft}}$, and kinematic consistency loss $\mathcal{L}_{\text{kine}}$. Performance is evaluated across six sensor dimensions using MAE and RMSE to assess absolute amplitude fidelity, together with R^2 and Pearson correlation to capture variance alignment and temporal agreement. This combination allows us to distinguish errors in signal scale from mismatches in motion dynamics.

4.7.1 Effect of Input Sources: Pose Guidance and Physics-Based Initialization. We first examine the role of PrimeIMU’s two complementary input sources. Removing the physics-based simulated IMU initialization (w/o phy imu) leads to a consistent and substantial degradation across both accelerometer and gyroscope channels. For accelerometers, errors increase sharply (e.g., Accel-X MAE increases from 0.0306 to 0.0722 and Pearson drops from 0.9886 to 0.8347), while gyroscopes exhibit pronounced scale mismatches (e.g., Gyro-X MAE increases from 3.05 to 4.88). These results indicate that the physics stage injects essential high-frequency and device-like inertial structure that cannot be reliably recovered from pose alone, particularly for angular velocity signals.

Removing pose guidance (w/o pose) produces a qualitatively different pattern. Across most axes, RMSE increases and Pearson correlation decreases slightly relative to the full model (e.g., Gyro-X Pearson 0.9944 vs. 0.9972; Accel-Z 0.9911 vs. 0.9916), suggesting weaker temporal alignment and less stable gravity anchoring. An isolated exception appears on Accel-X, where w/o pose yields lower MAE and RMSE (0.0235 vs. 0.0306) but also a lower Pearson correlation (0.9811 vs. 0.9886). This indicates a reduction in static bias on a single axis at the expense of temporal fidelity, a trade-off that does not generalize across channels. Since downstream inertial tasks rely primarily on correct motion dynamics rather than per-axis offset minimization, the full model remains preferable.

Overall, the two inputs serve distinct and complementary roles. Physics-based initialization provides realistic device-level dynamics and high-frequency structure, while pose guidance stabilizes global motion trends and gravity orientation. Using both jointly yields the strongest and most consistent performance across sensors and metrics.

4.7.2 Effect of Loss Functions: Reconstruction, Spectral Alignment, and Kinematic Consistency. We next analyze the contribution of individual loss terms, each of which targets a distinct failure mode in IMU synthesis. The reconstruction loss $\mathcal{L}_{\text{recon}}$ primarily controls amplitude and bias alignment in the time domain. Removing it causes the most severe degradation on gyroscopes, with MAE and RMSE increasing dramatically (e.g., Gyro-X MAE 9.44 vs. 3.05; Gyro-Z MAE 6.98 vs. 3.00) and R^2 dropping substantially, even though Pearson correlation remains relatively high. This reflects a characteristic failure mode in which temporal trends are preserved but signal magnitudes drift, underscoring the necessity of explicit time-domain supervision for sensor-faithful synthesis.

The spectral alignment loss $\mathcal{L}_{\text{stft}}$ shapes frequency content and suppresses non-physical oscillations introduced by differentiation. Without it, accelerometer errors increase markedly (e.g., Accel-X MAE 0.0745 vs. 0.0306; Accel-Z RMSE 0.0323 vs. 0.0262), accompanied by a clear drop in R^2 . Gyroscope channels also suffer increased scale error and reduced

Table 9. Ablation study of PrimeIMU on the UTD-MHAD dataset. We compare the full model with variants where key components are removed: pose sequence input (w/o pose), simulated IMU initialization from the physics stage (w/o phy imu), and individual loss terms for kinematic consistency ($\mathcal{L}_{\text{kine}}$), spectral alignment ($\mathcal{L}_{\text{stft}}$), and time-domain reconstruction ($\mathcal{L}_{\text{recon}}$). Results are reported across six sensor dimensions (Accel-X/Y/Z, Gyro-X/Y/Z) using MAE, RMSE, R^2 , and Pearson correlation. Bold denotes the best result within each sensor dimension, and underlines indicate the second-best.

Dimension	Model	MAE ↓	RMSE ↓	R^2 ↑	Pearson ↑
Accel-X	Full	0.0306	0.0399	0.8706	0.9886
	w/o pose	0.0235	0.0319	0.9323	<u>0.9811</u>
	w/o phy imu	0.0722	0.1118	0.3557	0.8347
	w/o $\mathcal{L}_{\text{kine}}$	<u>0.0295</u>	<u>0.0396</u>	<u>0.8902</u>	0.9762
	w/o $\mathcal{L}_{\text{stft}}$	0.0745	0.1117	0.3190	0.9494
	w/o $\mathcal{L}_{\text{recon}}$	0.0401	0.0520	0.1943	0.9708
Accel-Y	Full	0.0176	0.0230	0.9785	0.9977
	w/o pose	<u>0.0234</u>	<u>0.0326</u>	0.9585	0.9900
	w/o phy imu	0.0285	0.0384	0.9679	0.9933
	w/o $\mathcal{L}_{\text{kine}}$	0.0275	0.0373	0.9721	<u>0.9918</u>
	w/o $\mathcal{L}_{\text{stft}}$	0.0284	0.0366	<u>0.9725</u>	0.9922
	w/o $\mathcal{L}_{\text{recon}}$	0.0374	0.0475	<u>0.9349</u>	0.9912
Accel-Z	Full	0.0189	0.0262	0.9583	0.9916
	w/o pose	0.0276	0.0349	0.9343	<u>0.9911</u>
	w/o phy imu	0.0349	0.0459	0.8822	0.9822
	w/o $\mathcal{L}_{\text{kine}}$	0.0396	0.0496	0.8891	0.9763
	w/o $\mathcal{L}_{\text{stft}}$	<u>0.0250</u>	<u>0.0323</u>	0.9205	0.9884
	w/o $\mathcal{L}_{\text{recon}}$	0.0440	0.0559	0.7398	0.9741
Gyro-X	Full	3.0549	4.2513	0.9842	0.9972
	w/o pose	4.5961	5.7856	0.9462	0.9944
	w/o phy imu	4.8794	6.4089	0.9337	0.9910
	w/o $\mathcal{L}_{\text{kine}}$	<u>4.0421</u>	5.6858	0.9814	0.9926
	w/o $\mathcal{L}_{\text{stft}}$	4.1948	<u>5.5147</u>	<u>0.9750</u>	<u>0.9945</u>
	w/o $\mathcal{L}_{\text{recon}}$	9.4388	11.5457	0.8519	0.9862
Gyro-Y	Full	3.1581	4.0785	0.8006	0.9957
	w/o pose	<u>3.5308</u>	<u>4.2360</u>	0.8109	<u>0.9938</u>
	w/o phy imu	4.9996	6.6176	0.8989	0.9736
	w/o $\mathcal{L}_{\text{kine}}$	4.3202	5.9407	0.6626	0.9528
	w/o $\mathcal{L}_{\text{stft}}$	4.6992	5.9767	0.7854	0.9808
	w/o $\mathcal{L}_{\text{recon}}$	5.5411	7.0502	0.5790	0.9813
Gyro-Z	Full	2.9964	3.8424	0.8852	0.9983
	w/o pose	<u>4.3181</u>	<u>5.1476</u>	0.8355	<u>0.9965</u>
	w/o phy imu	5.2329	6.6543	0.8841	0.9926
	w/o $\mathcal{L}_{\text{kine}}$	4.8615	5.9656	0.8339	0.9935
	w/o $\mathcal{L}_{\text{stft}}$	5.1138	6.4306	0.9557	0.9906
	w/o $\mathcal{L}_{\text{recon}}$	6.9763	8.8849	0.8483	0.9863

correlation (e.g., Gyro-Z MAE 5.11 vs. 3.00). These results confirm that spectral supervision is critical for restoring IMU-like frequency characteristics that are not sufficiently constrained by time-domain losses alone.

The kinematic consistency loss $\mathcal{L}_{\text{kine}}$ enforces agreement between generated signals and pose-derived motion after integration. Disabling it disproportionately affects axes sensitive to accumulated drift and gravity alignment. For example, Accel-Z RMSE increases from 0.0262 to 0.0496 and Gyro-Z MAE rises from 3.00 to 4.86, with corresponding

reductions in correlation. This demonstrates that $\mathcal{L}_{\text{kine}}$ acts as an effective physical regularizer, preventing non-physical accelerations and long-term drift while allowing the model to learn fine-grained inertial detail.

4.7.3 Axis-Wise Behavior and Sensor-Specific Sensitivity. Examining axis-wise behavior reveals patterns consistent with inertial sensor physics. Accelerometer channels benefit most from the full composite objective, achieving near-ceiling correlations (e.g., Accel-Y Pearson 0.9977) and high variance explanation. They are particularly sensitive to the removal of spectral supervision and physics priors, both of which directly influence frequency content and impact responses.

Gyroscope channels, in contrast, maintain very high Pearson correlations across most ablations, indicating robust temporal alignment. However, they are significantly more sensitive to scale and bias errors when $\mathcal{L}_{\text{recon}}$ or $\mathcal{L}_{\text{stft}}$ is removed, as reflected by large increases in MAE and RMSE despite stable correlations. This recurring “high-correlation but wrong-scale” behavior motivates the inclusion of explicit amplitude- and spectrum-alignment terms and highlights the importance of reporting both correlation-based and error-based metrics.

4.7.4 Summary and Implications. While isolated ablations may yield marginal improvements on individual axes or metrics, the full PrimeIMU configuration consistently provides the strongest joint trade-off across MAE, RMSE, R^2 , and Pearson over all six sensor channels. The ablation results confirm that each component addresses a complementary aspect of IMU synthesis: physics priors inject device realism, pose guidance stabilizes global motion, reconstruction enforces correct scale, spectral alignment restores frequency structure, and kinematic consistency prevents drift. Together, these components produce sensor-faithful inertial signals that directly support the cross-dataset transfer and downstream action recognition gains demonstrated in Section 4.

5 Conclusion

We introduce PrimeIMU, a framework for synthesizing high-frequency, high-fidelity IMU signals directly from video. Our study demonstrates that PrimeIMU addresses three long-standing challenges in video-to-IMU generation: (i) error propagation from imperfect pose estimation, (ii) the loss of fine-grained motion dynamics caused by low video frame rates, and (iii) the mismatch between kinematic estimates and the measurements by physical IMU sensors. By combining physics-guided IMU simulation with a hybrid U-Net refinement module, PrimeIMU first converts video-derived kinematic trajectories into approximate inertial measurements and then learns to refine these simulated signals to more faithfully capture the dynamics and sensor characteristics of real IMU devices. Extensive experiments show that PrimeIMU (1) improves the fidelity of inertial signal synthesis, (2) generalizes to unseen motion patterns while preserving realistic signal characteristics, (3) enables synthetic-only training of downstream models with performance approaching that of real-sensor training and (4) provides additional gains when used to augment real data, and (5) adapts across datasets and sensor configurations with minimal fine-tuning, supporting robust downstream performance under domain shift.

References

- [1] Ghaith Al-refai, Dina Karasneh, Hisham Elmoaqet, Mutaz Ryalat, and Natheer Almtireen. 2025. Surface Classification from Robot Internal Measurement Unit Time-Series Data Using Cascaded and Parallel Deep Learning Fusion Models. *Machines* 13, 3 (2025), 251.
- [2] Lei Bai, Lina Yao, Xianzhi Wang, Salil S Kanhere, Bin Guo, and Zhiwen Yu. 2020. Adversarial multi-view networks for activity recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 2 (2020), 1–22.
- [3] Dmitrijs Balabka. 2019. Semi-supervised learning for human activity recognition using adversarial autoencoders. In *Adjunct proceedings of the 2019 ACM international joint conference on pervasive and ubiquitous computing and proceedings of the 2019 ACM international symposium on wearable computers*. 685–688.
- [4] Chen Chen, Roozbeh Jafari, and Nasser Kehtarnavaz. 2015. UTD-MHAD: A multimodal dataset for human action recognition utilizing a depth camera and a wearable inertial sensor. In *International Conference on Image Processing*.
- [5] Wenqiang Chen, Shupeí Lin, Elizabeth Thompson, and John Stankovic. 2021. Sensecollect: We need efficient ways to collect on-body sensor-based human activity data! *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 3 (2021), 1–27.
- [6] Dongzhou Cheng, Lei Zhang, Can Bu, Xing Wang, Hao Wu, and Aiguo Song. 2023. ProtoHAR: Prototype guided personalized federated learning for human activity recognition. *IEEE Journal of Biomedical and Health Informatics* 27, 8 (2023), 3900–3911.
- [7] Giovanni Cioffi, Leonard Bauersfeld, Elia Kaufmann, and Davide Scaramuzza. 2023. Learned inertial odometry for autonomous drone racing. *IEEE Robotics and Automation Letters* 8, 5 (2023), 2684–2691.
- [8] Asiye Demirtaş, Gökhan Erdemir, and Haluk Bayram. 2024. Indoor surface classification for mobile robots. *PeerJ Computer Science* 10 (2024), e1730.
- [9] Siwei Feng and Marco F Duarte. 2019. Few-shot learning-based human activity recognition. *Expert Systems with Applications* 138 (2019), 112782.
- [10] Iuri Frosio, Federico Pedersini, and N Alberto Borghese. 2008. Autocalibration of MEMS accelerometers. *IEEE Transactions on instrumentation and measurement* 58, 6 (2008), 2034–2041.
- [11] Walid Gomaa and Mohamed A Khamis. 2023. A perspective on human activity recognition from inertial motion data. *Neural Computing and Applications* (2023).
- [12] Harish Haresamudram, Apoorva Beedu, Varun Agrawal, Patrick L Grady, Irfan Essa, Judy Hoffman, and Thomas Plötz. 2020. Masked reconstruction based self-supervision for human activity recognition. In *Proceedings of the 2020 ACM International Symposium on Wearable Computers*. 45–49.
- [13] Hyeokhyen Kwon, Catherine Tong, Harish Haresamudram, Yan Gao, Gregory D Abowd, Nicholas D Lane, and Thomas Ploetz. 2020. Imutube: Automatic extraction of virtual on-body accelerometry from video for human activity recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (2020).
- [14] Arttu Lämsä, Jaakko Tervonen, Jussi Liikka, Constantino Álvarez Casado, and Miguel Bordallo López. 2022. Video2IMU: Realistic IMU features and signals from videos. In *International Conference on Wearable and Implantable Body Sensor Networks*.
- [15] Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, and Thomas Plötz. 2024. Imugpt 2.0: Language-based cross modality transfer for sensor-based human activity recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (2024).
- [16] Zikang Leng, Hyeokhyen Kwon, and Thomas Ploetz. 2023. Generating Virtual On-Body Accelerometer Data from Virtual Textual Descriptions for Human Activity Recognition. In *International Symposium on Wearable Computers*.
- [17] Faisal Mohd-Yasin, Can E Korman, and David J Nagel. 2003. Measurement of noise characteristics of MEMS accelerometers. *Solid-State Electronics* 47, 2 (2003), 357–360.
- [18] Andreas Mueller. 2019. Modern robotics: Mechanics, planning, and control [bookshelf]. *IEEE Control Systems Magazine* 39, 6 (2019), 100–102.
- [19] Bartłomiej Nalepa, Magdalena Pawlyta, Mateusz Janiak, Agnieszka Szczęsna, Aleksander Gwiazda, and Konrad Wojciechowski. 2022. Recreating the Motion Trajectory of a System of Articulated Rigid Bodies on the Basis of Incomplete Measurement Information and Unsupervised Learning. *Sensors* 22, 6 (2022), 2198.
- [20] Francisco Javier Ordóñez and Daniel Roggen. 2016. Deep convolutional and lstm recurrent neural networks for multimodal wearable activity recognition. *Sensors* (2016).
- [21] Aakash Parmar, Rakesh Katariya, and Vatsal Patel. 2018. A review on random forest: An ensemble classifier. In *International Conference on Intelligent Data Communication Technologies and Internet of Things*.
- [22] Dario Pavlo, Christoph Feichtenhofer, David Grangier, and Michael Auli. 2019. 3d human pose estimation in video with temporal convolutions and semi-supervised training. In *Conference on Computer Vision and Pattern Recognition*.
- [23] Thomas Plötz. 2023. If only we had more data!: Sensor-based human activity recognition in challenging scenarios. In *2023 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*. IEEE, 565–570.
- [24] Sathian Pookkuttath, Povendhan Arthanaripalayam Palanisamy, and Mohan Rajesh Elara. 2023. AI-Enabled Condition Monitoring Framework for Outdoor Mobile Robots Using 3D LiDAR Sensor. *Mathematics* 11, 16 (2023), 3594.
- [25] Yuantian Qin, Zhehang Yin, Quanou Yang, and Kai Zhang. 2024. Dynamics Parameter Identification of Articulated Robot. *Machines* 12, 9 (2024), 595.
- [26] Yuheng Qiu, Can Xu, Yutian Chen, Shibo Zhao, Junyi Geng, and Sebastian Scherer. 2025. AirIO: Learning Inertial Odometry with Enhanced IMU Feature Observability. *arXiv preprint arXiv:2501.15659* (2025).
- [27] Daniele Ravi, Charence Wong, Benny Lo, and Guang-Zhong Yang. 2016. Deep learning for human activity recognition: A resource efficient implementation on low-power devices. In *2016 IEEE 13th international conference on wearable and implantable body sensor networks (BSN)*. IEEE,

- 71–76.
- [28] Aaqib Saeed, Tanir Ozcelebi, and Johan Lukkien. 2019. Multi-task self-supervised learning for human activity detection. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 2 (2019), 1–30.
 - [29] Panneer Selvam Santhalingam, Parth Pathak, Huzefa Rangwala, and Jana Kosecka. 2023. Synthetic smartwatch imu data generation from in-the-wild asl videos. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (2023).
 - [30] Allah Bux Sargano, Xiaofeng Wang, Plamen Angelov, and Zulfiqar Habib. 2017. Human action recognition using transfer learning with deep representations. In *2017 International joint conference on neural networks (IJCNN)*. IEEE, 463–469.
 - [31] Ronald W Schafer. 2011. What is a savitzky-golay filter?[lecture notes]. *IEEE Signal Processing Magazine* (2011).
 - [32] Satya P Singh, Madan Kumar Sharma, Aimé Lay-Ekuakille, Deepak Gangwar, and Sukrit Gupta. 2020. Deep ConvLSTM with self-attention for human activity decoding using wearable sensors. *IEEE Sensors Journal* (2020).
 - [33] David Strömback, Sangxia Huang, and Valentin Radu. 2020. Mm-fit: Multimodal deep learning for automatic exercise logging across sensing devices. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (2020).
 - [34] Ruslan Sultan and Steffen Greiser. 2025. Time and Frequency Domain Analysis of IMU-based Orientation Estimation Algorithms with Comparison to Robotic Arm Orientation as Reference. *Sensors* 25, 16 (2025), 5161.
 - [35] Shan Suthaharan. 2016. Support vector machine. In *Machine learning models and algorithms for big data classification: thinking with examples for effective learning*. Springer.
 - [36] Nilay Tufek, Murat Yalcin, Mucahit Altintas, Fatma Kalaoglu, Yi Li, and Senem Kursun Bahadir. 2019. Human action recognition using deep learning methods on limited sensory data. *IEEE Sensors Journal* (2019).
 - [37] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in Neural Information Processing Systems* (2017).
 - [38] Gregory W Vogl, M Alkan Donmez, Andreas Archenti, and Brian A Weiss. 2016. Inertial measurement unit for on-Machine diagnostics of machine tool linear axes. In *Annual Conference of the PHM Society*, Vol. 8.
 - [39] Cong Xu, Yuhang Li, Dae Lee, Dae Hoon Park, Hongda Mao, Huyen Do, Jonathan Chung, and Dinesh Nair. 2023. Augmentation robust self-supervised learning for human activity recognition. In *International Conference on Acoustics, Speech and Signal Processing*.
 - [40] Yufei Xu, Jing Zhang, Qiming Zhang, and Dacheng Tao. 2023. Vitpose++: Vision transformer for generic body pose estimation. *Transactions on Pattern Analysis and Machine Intelligence* (2023).
 - [41] Xiyuan Zhang, Diyan Teng, Ranak Roy Chowdhury, Shuheng Li, Dezhi Hong, Rajesh Gupta, and Jingbo Shang. 2024. UniMTS: Unified Pre-training for Motion Time Series. In *Advances in Neural Information Processing Systems*.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009